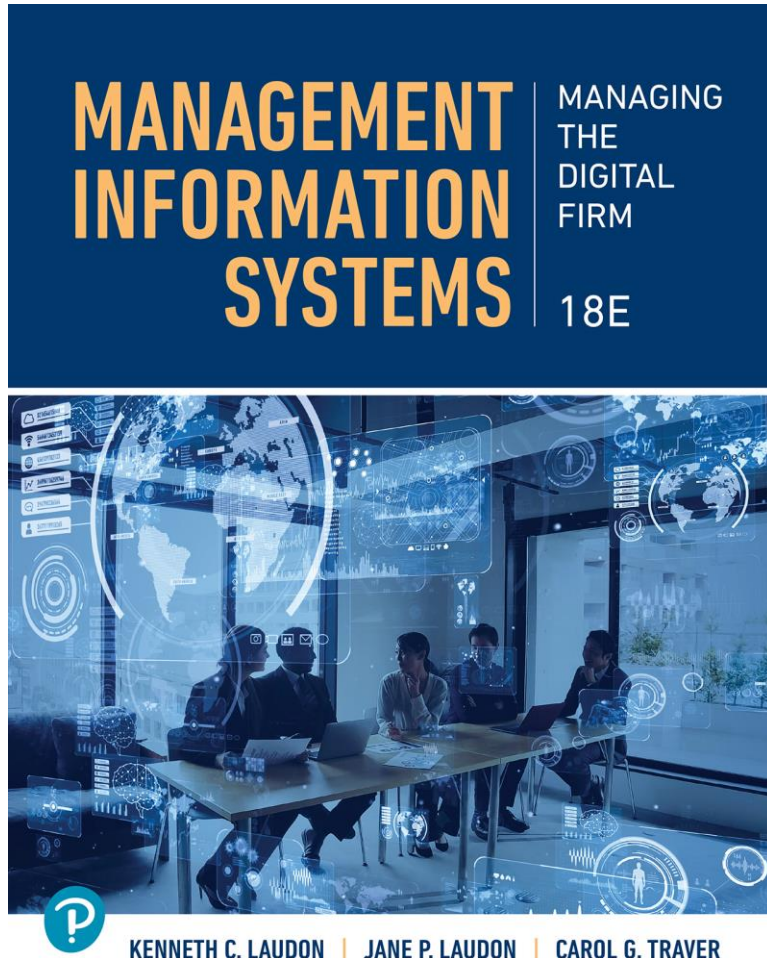


Management Information Systems: Managing the Digital Firm

Eighteenth Edition



Chapter 14

Making the Business Case for Information Systems and Managing Projects

Learning Objectives

- 14.1** Explain how to build a business case.
- 14.2** Describe methods for evaluating IT investments.
- 14.3** Describe project management methodologies and variables.
- 14.4** Understand the importance of effective project management.
- 14.5** Discuss project risk factors.
- 14.6** Understand how project risks can be managed.
- 14.7** Understand how the information in this chapter can help your career.

How to Build a Business Case (1 of 3)

- Project
 - A planned series of related activities for achieving a specific business objective
- Business case
 - A proposal to management seeking approval for an investment
 - A business case for an IT investment describes the problem facing the organization that can be solved by investing in a proposed system solution

How to Build a Business Case (2 of 3)

- Important elements of a business case (a proposal to management seeking approval for an investment) are
 - Describes problem facing the organization
 - Provides analysis of costs, benefits, and risks
 - Describes rationale for proceeding with investment
 - Shows how investment supports firm's strategic goals

How to Build a Business Case (3 of 3)

- A business case
 - Provides information necessary to make informed decision
 - Explains how investment will provide value for business
 - Identifies risks and alternative solutions
 - Describes how solution may require changes in organizational culture, systems, processes, jobs

Figure 14.1 Factors to Consider in Making the Business Case



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The Information Systems Plan (1 of 2)

- Information systems plan
 - Serves as a road map indicating the direction of systems development
 - Purpose of the plan
 - Strategic business plan rationale
 - Current systems
 - New developments
 - Management strategy
 - Implementation plan
 - Budget

The Information Systems Plan (2 of 2)

- To plan effectively
 - Inventory and document applications, infrastructure components, and long- and short-term information requirements
 - Identify the decision improvements that would provide the greatest additional value to the firm
 - Then develop metrics to quantify the value of more timely and precise information on the outcome of the decision

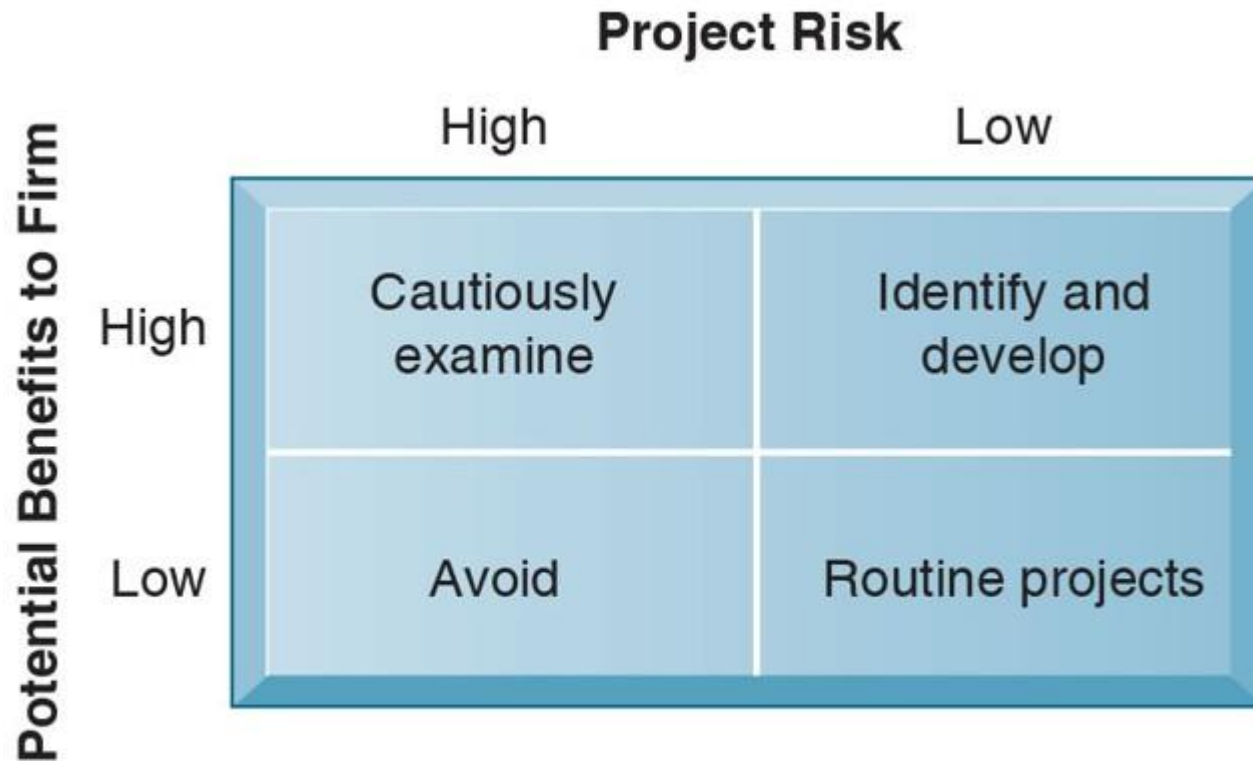
Methods for Evaluating IT Investments

- There are a variety of methods that can be used to evaluate a proposed IT investment
 - Portfolio analysis
 - Scoring models
 - Capital budgeting models
 - Software tools

Portfolio Analysis

- Portfolio analysis
 - A tool to evaluate alternative systems projects
 - Inventories all of an organization's information systems projects and assets
- Each system has a profile of risk and benefit
 - High benefit, low risk
 - High benefit, high risk
 - Low benefit, low risk
 - Low benefit, high risk
- To improve return on portfolio, balance risk and return from systems investments

Figure 14.2 A System Portfolio



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Scoring Models

- Scoring model
 - Useful for selecting projects where many criteria must be considered
 - Assigns weights to various features of system and calculates weighted totals
 - Many qualitative judgments involved
 - Requires experts who understand the issues and the technology

Determining Solution Costs and Benefits (1 of 5)

- Information system costs and benefits
 - Tangible benefits
 - Can be quantified and assigned a monetary value
 - Intangible benefits
 - Cannot be immediately quantified but may lead to quantifiable gains in the long run

Determining Solution Costs and Benefits (2 of 5)

- To determine the benefits of a particular project, you'll need to calculate all of its costs and all of its benefits
- Capital budgeting models are one way to measure the value of investing in long-term information systems projects
 - Measure value of long-term capital investment projects
 - Rely on measures of the firm's cash outflows and inflows

Determining Solution Costs and Benefits (3 of 5)

- Principal capital budgeting models for evaluating IT projects are
 - Payback method
 - Accounting rate of return on investment (ROI)
 - Net present value (NPV), and
 - Internal rate of return (IRR)

Determining Solution Costs and Benefits (4 of 5)

- Limitations of financial models
 - Many companies' information systems investment decisions do not adequately consider costs from organizational disruptions created by a new system
 - Train end users
 - Impact that users' learning curves for a new system have on productivity
 - Time managers need to spend overseeing new system-related changes

Determining Solution Costs and Benefits (5 of 5)

- Software to help make technology investment decisions
 - Technology business management (TBM)
 - A framework aimed at improving business outcomes
 - By giving organizations a consistent method of translating technology investments
- IT financial management tools (ITEM tools)
 - Automate TBM and help organizations analyze, plan, and control costs

Project Management Methodologies and Variables (1 of 3)

- Project management
 - Application of knowledge, skills, tools, and techniques to achieve specific targets within specified budget and time constraints
 - Planning the work, assessing risk, estimating resources, organizing the work, acquiring human and material resources, assigning tasks, directing activities, controlling project execution, reporting progress, and analyzing the results

Project Management Methodologies and Variables (2 of 3)

- Waterfall project management approach
 - Traditional format for project management
 - A somewhat rigid and linear approach
- Agile project management
 - Allows for more adaptability and flexibility
 - Uses “sprints” where the project team works to complete a set amount of work within a short, specified time period

Project Management Methodologies and Variables (3 of 3)

- Project management variables
 - Scope
 - Time
 - Cost
 - Quality
 - Risk
- Technical debt (also called tech debt or code debt)
 - Results from the trade-offs between time, cost, and quality that often occur in the development of an information system

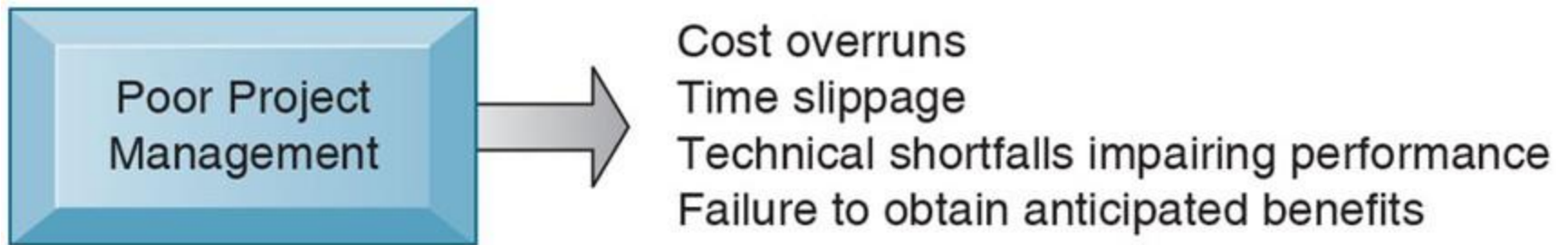
Importance of Effective Project Management (1 of 3)

- Very high failure rate among information systems projects that have not been properly managed
- In nearly every organization, information systems projects take much more time and money to implement than originally anticipated
 - Or the completed system does not work properly

Importance of Effective Project Management (2 of 3)

- How badly are projects managed?
 - On average, private sector projects are underestimated by half in terms of budget and time required to deliver the complete system promised in the system plan
- A systems development project without proper management will most likely suffer any or all of the following
 - Costs that vastly exceed budgets
 - Unexpected time slippage
 - Technical performance that is less than expected
 - Failure to obtain anticipated benefits

Figure 14.4 Consequences of Poor Project Management



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Importance of Effective Project Management (3 of 3)

- The way in which nontechnical business users must interact with a system may be excessively complicated and discouraging
- A system maybe designed with a poor user interface
 - User interface is the part of the system with which end users interact

Project Risk Factors (1 of 5)

- Systems differ dramatically in their size, scope, level of complexity, and organizational and technical components
- Dimensions of project risk may include
 - Project size
 - Project structure
 - Experience with technology

Project Risk Factors (2 of 5)

- Change management
 - Management of the organizational change associated with an innovation, such as the introduction of a new system
- Implementation
 - All organizational activities working toward the adoption, management, and routinization of an innovation, such as a new information system
- Change agent
 - Catalyst for the entire change process and responsible for ensuring that all parties involved accept the changes created by a new system.

Project Risk Factors (3 of 5)

- Role of end users
 - With high levels of user involvement
 - More opportunities to mold the system according to their priorities
 - More likely to react positively to the completed system
- User-designer communication gap
 - Users and information systems specialists tend to have different backgrounds, interests, and priorities

Project Risk Factors (4 of 5)

- Management support and commitment
 - Affects positive perception by both users and technical staff
 - Ensures sufficient funding and resources
 - Helps enforce required organizational changes

Project Risk Factors (5 of 5)

- Change management challenges for business process management (BPM), enterprise applications, and mergers and acquisitions
 - Very high failure rate among enterprise application and BPM projects (up to 70 percent for far-reaching BPM)
 - Poor implementation and change management practices
 - Mergers and acquisitions
 - Similarly high failure rate of integration projects
 - Merging of systems of two companies requires:
 - Considerable organizational change, complex systems projects

How Project Risks Can Be Managed

(1 of 2)

- First step involves identifying nature and level of risk of a project
- Each project can then be managed with tools and risk-management approaches geared to level of risk

How Project Risks Can Be Managed

(2 of 2)

- Managing technical complexity
 - Internal integration tools
 - Project leaders with technical and administrative experience
 - Highly experienced team members
 - Frequent team meetings
 - Securing of technical experience outside firm if necessary

Formal Planning and Control Tools

(1 of 2)

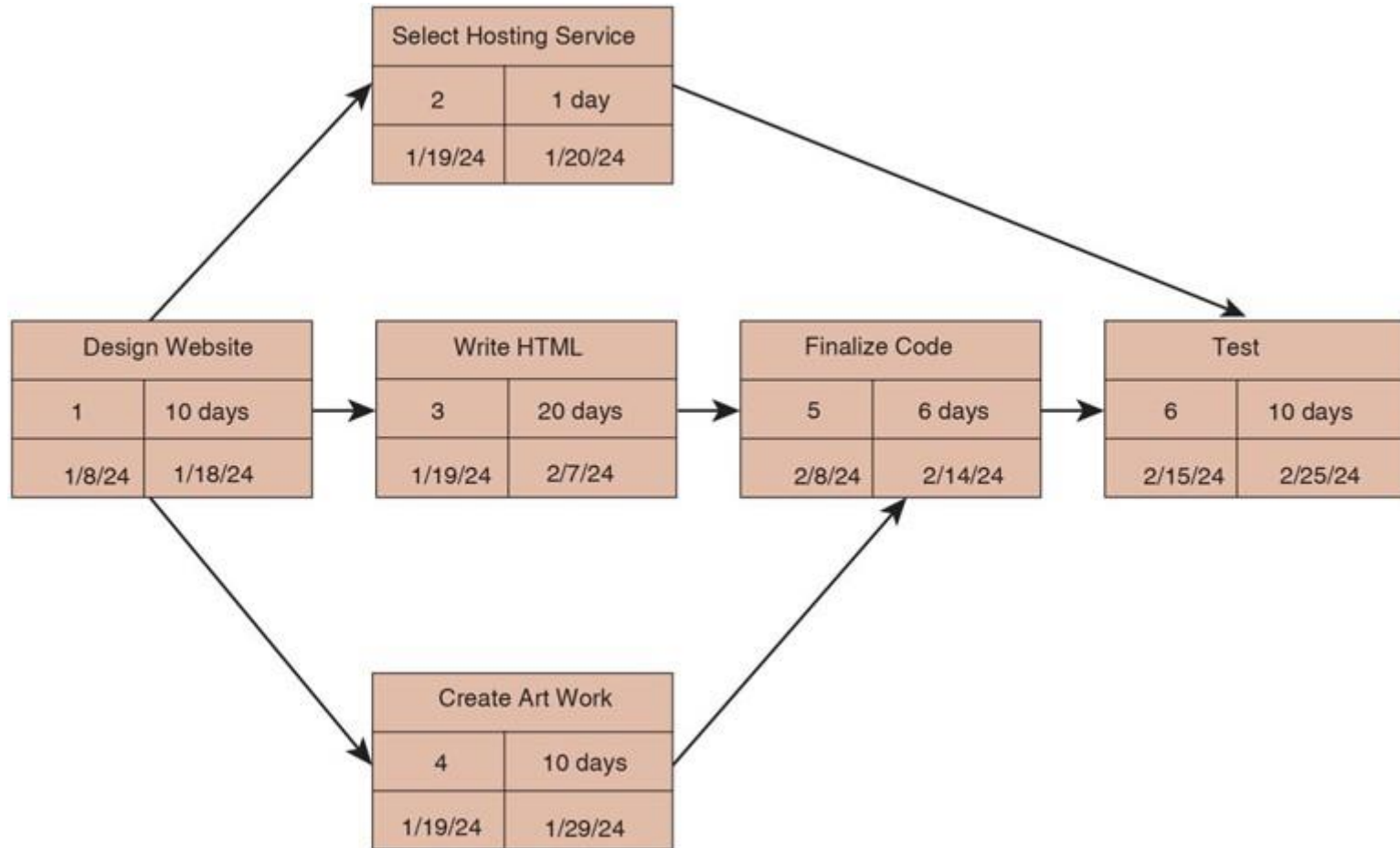
- Used for documenting and monitoring project plans
- Help identify bottlenecks and impact of problems
- Gantt chart
 - Visual representation of timing and duration of tasks
 - Human resource requirements of tasks

Formal Planning and Control Tools

(2 of 2)

- PERT charts
 - Graphically depict tasks and interrelationships
 - Indicate sequence of tasks necessary
- Kanban board
 - A visual tool that uses “cards” to represent specific tasks, and columns to represent each stage of the project process
 - Often used as a part of agile project management

Figure 14.6 A Pert Chart



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Increasing User Involvement and Overcoming User Resistance (1 of 2)

- External integration tools
 - Link work of implementation team to users at all levels
- User resistance to organizational change
 - Counterimplementation
 - A deliberate strategy to thwart the implementation of an information system or an innovation in an organization

Increasing User Involvement and Overcoming User Resistance (2 of 2)

- Strategies to overcome user resistance
 - User participation, education and training
 - Management edicts and policies
 - Incentives for cooperation
 - Improvement of end-user interface
 - Resolution of organizational problems prior to introduction of new system

Designing for the Organization

- Need to address ways in which organization changes with new system
 - Ergonomics
 - Interaction of people and machines in work environment
 - Organizational impact analysis
 - How system will affect organizational structure, attitudes, decision making, operations
 - Sociotechnical design
 - Addresses human and organizational issues

Project Management Software Tools

- Can automate many aspects of project management
- Capabilities for defining, ordering tasks
 - Assigning resources to tasks, tracking progress
 - Manage very large numbers of tasks and relationships
- Microsoft Project
- Cloud-based software
- Project portfolio management software (PPM software)
 - Helps organizations manage portfolios of projects and dependencies among them

AI and Project Management Software Tools

- Project management software tools are increasingly incorporating AI functionality to enhance efficiency and project workflow.
- AI tools can help automate routine project management tasks
 - Data entry
 - Scheduling
 - Creating meeting summaries
 - Sending notifications
 - Updating project status

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